

Tech & Teach

Digital Chip Design Program

RTL Design

Task #10

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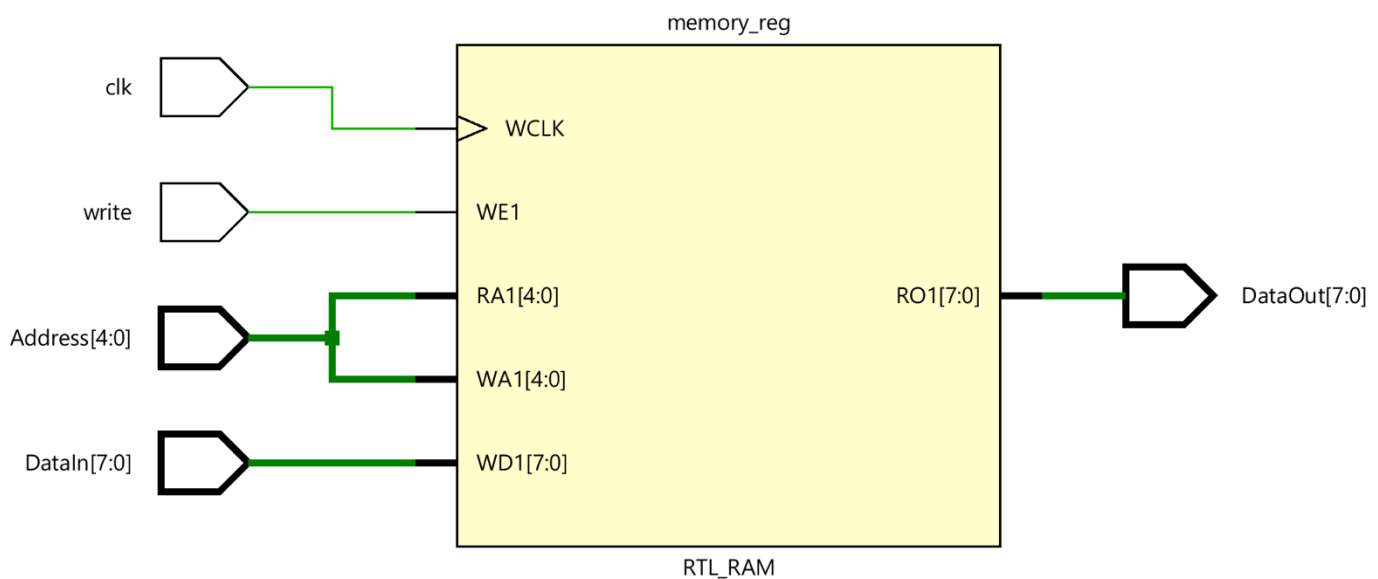
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Seq & Comb Circuits Datapath – Part 1

Version 1.0

Task #1.1

```
1 module SRAM
2   #(parameter Width = 8, Depth = 32 , AddrWidth = $clog2(Depth))
3   (
4     input clk, write,
5     input logic [AddrWidth-1:0] Address ,
6     input logic [Width-1:0] DataIn,
7     output logic [Width-1:0] DataOut
8   );
9
10  logic [Width-1:0] memory [0:Depth-1];
11
12  assign DataOut = memory[Address];
13
14  always@(posedge clk) begin
15    if (write)
16      memory[Address] <= DataIn;
17  end
18 endmodule
```



```

1  `timescale 1ns / 1ps
2  module SRAM_tb
3      #(parameter Width = 8, Depth = 32 , AddrWidth = $clog2(Depth)) ();
4
5      logic clk, write;
6      logic [AddrWidth-1:0] Address ;
7      logic [Width-1:0] DataIn;
8      logic [Width-1:0] DataOut;
9
10     SRAM #(
11         .Width(Width),
12         .Depth(Depth)
13     ) dut (
14         .clk(clk),
15         .write(write),
16         .Address(Address),
17         .DataIn(DataIn),
18         .DataOut(DataOut)
19     );
20
21     always #5 clk = ~clk;
22
23     initial begin
24         clk = 0;
25         write = 0;
26         Address = 0;
27         DataIn = 0;
28
29         @(negedge clk);
30         write = 1;
31         Address = 1;
32         DataIn = 8'hA5;
33
34         @(negedge clk);
35         write = 0;
36         Address = 0;
37
38         #6
39         Address = 1;
40
41         @(negedge clk);
42         write = 1;
43         Address = 2;
44         DataIn = 8'hFF;
45
46         @(negedge clk);
47         write = 0;
48         Address = 2;
49
50         #50 $finish;
51     end
52 endmodule

```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1	✓	✓	✓